

Quantum Computing

1 Exam Exercises

Exercise 1.1 Consider the state $|\psi\rangle = \frac{2+i}{3}|0\rangle - \frac{2}{3}|1\rangle$. If we measure $|\psi\rangle$ in the base $\{|u\rangle = \frac{i+1}{2}|0\rangle + \frac{i-1}{2}|1\rangle, |u^\dagger\rangle = \frac{i+1}{2}|0\rangle - \frac{i-1}{2}|1\rangle\}$, what is the probability of obtaining $|u\rangle$?

Exercise 1.2 Construct the two-qubit state $|a, b\rangle$, where $a = \frac{3}{5}|0\rangle + \frac{4}{5}|1\rangle$ and $b = \frac{1}{\sqrt{2}}|0\rangle + \frac{i}{\sqrt{2}}|1\rangle$.

Exercise 1.3 Which of the following states is entangled?

- a) $|++\rangle$
- b) $\frac{4}{5}|01\rangle - \frac{3}{5}|11\rangle$
- c) $\frac{1}{\sqrt{3}}|01\rangle + \frac{1}{\sqrt{3}}|10\rangle - \frac{1}{\sqrt{3}}|11\rangle$
- d) $\frac{1+i}{2\sqrt{2}}|++\rangle - \frac{1+i}{2\sqrt{2}}|+-\rangle + \frac{1-i}{2\sqrt{2}}|-+\rangle - \frac{1-i}{2\sqrt{2}}|--\rangle$

Exercise 1.4 Consider the unitary matrix U corresponding to the circuit in Figure 1.

- a) Calculate the inner product between rows 3 and 14 of U .
- b) Calculate the inner product of 3 with itself.

Exercise 1.5 Given a quantum circuit transforming $|0\rangle$ into $|+\rangle$ and $|1\rangle$ into $|-\rangle$, write down its output on input $\frac{i\sqrt{2}}{\sqrt{3}}|+\rangle + \frac{1}{\sqrt{3}}|-\rangle$.

Exercise 1.6 The circuit in Figure 2 is a test for $|u\rangle = |v\rangle$.

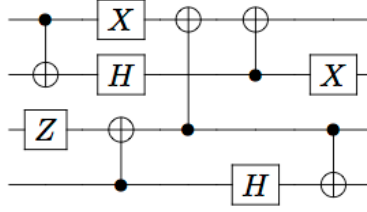


Figure 1: Un circuito su 4 qubit

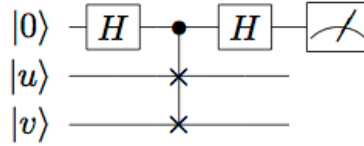


Figure 2: Un circuito che fa il test $|u\rangle = |v\rangle$

- a) Is $|u\rangle = |v\rangle = |+\rangle$, what is the probability of measuring 0?
- b) If $|u\rangle = |+\rangle$ and $|v\rangle = |-\rangle$, what is the probability of measuring 0?

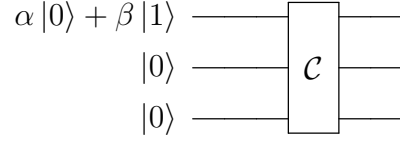
Exercise 1.7 Alice e Bob share the state $a|++\rangle + b|--\rangle$ with the first qubit owned by Alice and the second by Bob. Alice measures her qubit in the standard base and send it to Bob. Which operations among X, Z, H e I must apply Bob to obtain the state $a|0\rangle + b|1\rangle$, in each of the two cases below?

- Alice's measurements results in 0,
- Alice's measurements results in 1.

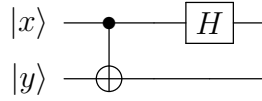
Exercise 1.8 Alice wants to send to Bob two qubit in the Bell state $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$, by quantum teleporting each qubit separately.

- a) what is the state of the two qubits received by Bob?
- b) How many classical bits must be sent to Bob?

Exercise 1.9 Describe a possible implementation of the circuit $\mathcal{C}$ in the figure below, so as to achieve the result $\alpha|000\rangle + \beta|111\rangle$ on the given input.



Exercise 1.10 For the circuit



find out the preparation of the input $|x\rangle|y\rangle$ which leads to the output

- a) $|00\rangle$,
- b) $|11\rangle$,
- c) $\frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$,

respectively.

Exercise 1.11 Compute the result of applying the Hadamard transform $H^{\otimes n}$ to the state

$$\frac{1}{\sqrt{2}}(|0\rangle^{\otimes n} + |1\rangle^{\otimes n}).$$

Exercise 1.12 Determine $|\phi\rangle$ such that $H^{\otimes 3}|\phi\rangle = \frac{1}{\sqrt{2}}(|000\rangle + |111\rangle)$.

Exercise 1.13 Let $|\phi\rangle = \sum_{y \in \{0,1\}^n} \beta_y |y\rangle$ be a state such that $\beta_y = 0$ if $s \cdot y = 1 \pmod{2}$ and $\beta_y = \frac{1}{\sqrt{2^n-1}}$ if $s \cdot y = 0 \pmod{2}$, where s is a string different from $00 \dots 0$.

- a) What is the probability of achieving s by performing a Fourier sampling (i.e. Hadamard followed by a measurement on the standard basis) on $|\phi\rangle$?
- b) What is the probability of achieving a result different from s ?